COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 1 – SPRING 2026

Course Title:	Digital System Design			Course Code:	CPE344	Credit Hours: 4(3,1)
Instructor:	Dr Muhammad Babar Ali			Office:	B-4	
Batch-Section:	FA23-B	Semester:	6	Program Name:	BSc in Computer Engineering	
Chapters/ Topics:	Ch 2-4			**Max. Marks:**	**40**	**Due Date:**
Student's Name:	**Muhammad Ahmad**			Reg. No.	**FA23 - BCE - 113**	

Important Instructions / Guidelines:

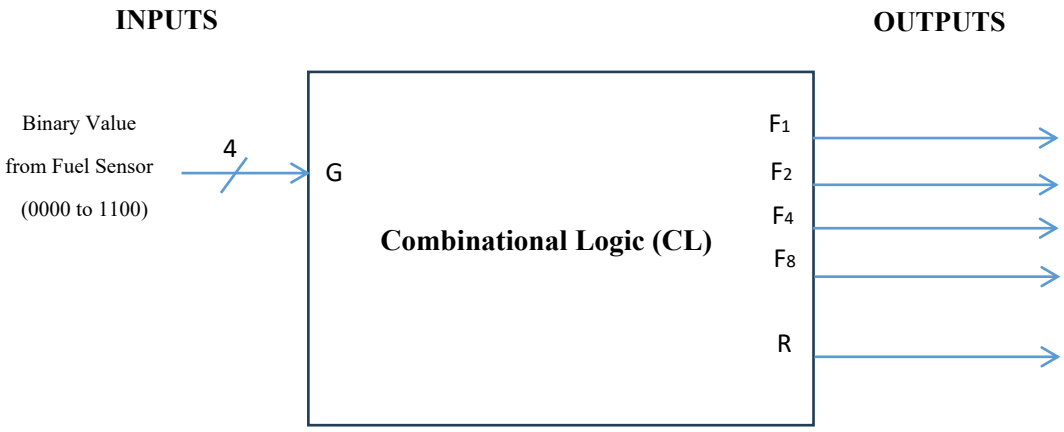
- Do not give multiple answers to a question.
- Do not use more than one ink color. Use only blue or black fountain pen or ballpoint pen.
- Plagiarism of work from any source will be penalized and may result in zero out of 40 marks in the assessment.
- Submit handwritten assignment on loose pages. Do not use pulled/ripped out papers from notebook.

Question 1: **(Marks 20)**

A vehicle has a fuel sensor which outputs the current fuel/gasoline level as a 4-bit binary number ranges from 0001_2 (0_{10}) to 1100_2 (12_{10}). Design combinational logic (CL) circuit shown in figure below that takes sensor data as input G and generates F1, F2, F4, F8 and R signals on its outputs for fuel indicators accordingly as described below:

- When Gasoline is above 11, CL sets all F_i outputs (where i =1, 2, 4, and 8) to 1 except R which is cleared.
- When Gasoline drops to 11 or below (and stays above 75%), CL sets outputs F_1, F_2 and F_4 to 1.
- When Gasoline drops to 75% or below (and stays above 50%), CL sets outputs F_1 and F_2 to 1
- When Gasoline drops to 50% or below (and stays above 25%), CL sets only output F_1 to 1
- When Gasoline drops to 25% or below, CL clears all outputs to 0
- When the fuel tank level is below 2, CL sets output R only to 1 and clears all F_i outputs (where i =1, 2, 4, and 8) outputs

Implement logic diagram of combinational circuit with five 2 × 4 decoders with enable input.



Question 2: **(Marks 10)**

You have been assigned to design a digital controller (finite state machine) for an automatic photo booth machine installed in a department lounge. The cost of taking a photo is 25 cents, as part of the cost is subsidized by the IEEE student branch. The machine accepts nickels, dimes, and quarters i.e., 5, 10 and 25 cents respectively. When enough coins have been inserted, the machine captures and prints the photo (assuming you are ready for photo in front of embedded camera in the machine) and returns any required change. The FSM inputs are N, D, and Q to respectively indicate Nickel, Dime, and Quarter coin were inserted. Assume that exactly one coin is inserted on each cycle. The outputs are P, R_N, R_D, and R_T to respectively indicate Print Photo, Return Nickel, Return Dime, and Return Two Dimes. When the total inserted amount reaches or exceeds 25 cents, the FSM should trigger the photo capture and return the appropriate change if necessary. After completing the operation, the machine must reset to its initial state and be ready to accept coins for the next user.

Question 3: **(Marks 10)**

The finite state machine has one input S and three outputs x, y, and z. The xyz should always exhibit the following sequence: 000, 001, 010, 100, repeat. The output should change only on a rising clock edge. Make 000 the initial state. Draw FSM for each of the following part:

a) An input S can stop the output sequence when set to 1. After that, when input S returns to 0 again, the output sequence also starts from 000 again. Design and implement the FSM controller that applies this behavior.
b) An input S postpones the output sequence for at least three cycles when set to 1. After these (halt–the–sequence) cycles, when input S returns to 0 again, the output sequence continues to the next value. Design the FSM controller that applies this behavior.
c) An input S postpones the output sequence for at least 30 cycles when set to 1. After these (halt–the–sequence) cycles, when input S returns to 0 again, the output sequence continues to the next value. Design the FSM controller that applies this behavior. Use appropriate MSI component to keep FSM simple.

Good Luck ☺

FA23-BCE-113

Q 1:

Given:

fuel sensor output 4 bit binary number [0000_2 - 1111_2]

combinational logic circuit

implement using 5 [2×4 decoder].

Input:

$G = G_3 G_2 G_1 G_0$

Output:

F_1, F_2, F_4, F_8 (fuel level indicator)

R (reserve warning light)

Conditions

① If $G > 1011_2$

$F_1 = 1, F_2 = 1, F_4 = 1, F_8 = 1$

$R = 0$

② If $G \leq 1011_2$ & $G \geq 1001_2$

$F_1 = 1, F_2 = 1, F_4 = 1, F_8 = 0$

$R = 0$

③ If $G \leq 1001_2$ & $G \geq 0110_2$

$F_1 = 1, F_2 = 1, F_4 = 0, F_8 = 0$

$R = 0$

④ If $G \leq 0110_2$ & $G \geq 0011_2$

$F_1 = 1, F_2 = 0, F_4 = 0, F_8 = 0$

$R = 0$

⑤ If $G \leq 0011_2$ & $G \geq 0010_2$

$F_1 = 0, F_2 = 0, F_4 = 0, F_8 = 0$

$R = 0$

If $G < 0010_2$

$F_1 = 0, F_2 = 0, F_3 = 0, F_4 = 0$

$R = 1$

Truth Table:

Input				Outputs				
G_3	G_2	G_1	G_0	F_1	F_2	F_4	F_8	R
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0
0	1	0	0	1	0	0	0	0
0	1	0	1	1	0	0	0	0
0	1	1	0	1	0	0	0	0
0	1	1	1	1	1	0	0	0
1	0	0	0	1	1	0	0	0
1	0	0	1	1	1	0	0	0
1	0	1	0	1	1	1	0	0
1	0	1	1	1	1	1	0	0
1	1	0	0	1	1	1	1	0
1	1	0	1	1	1	1	1	0
1	1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	1	0

Reserve bet

Derive boolean Equations:

(using K-map.

① F_8

G_3G_2 \ G_1G_0	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	0	0

$F_8 = G_3G_2$

② F_4

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

$F_4 = G_3G_2 + G_3G_1$

$F_4 = G_3(G_2 + G_1)$

③ F_2

	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	1	1	1	1
10	1	1	1	1

$F_2 = G_3 + G_2G_1G_8$

④ F_1

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

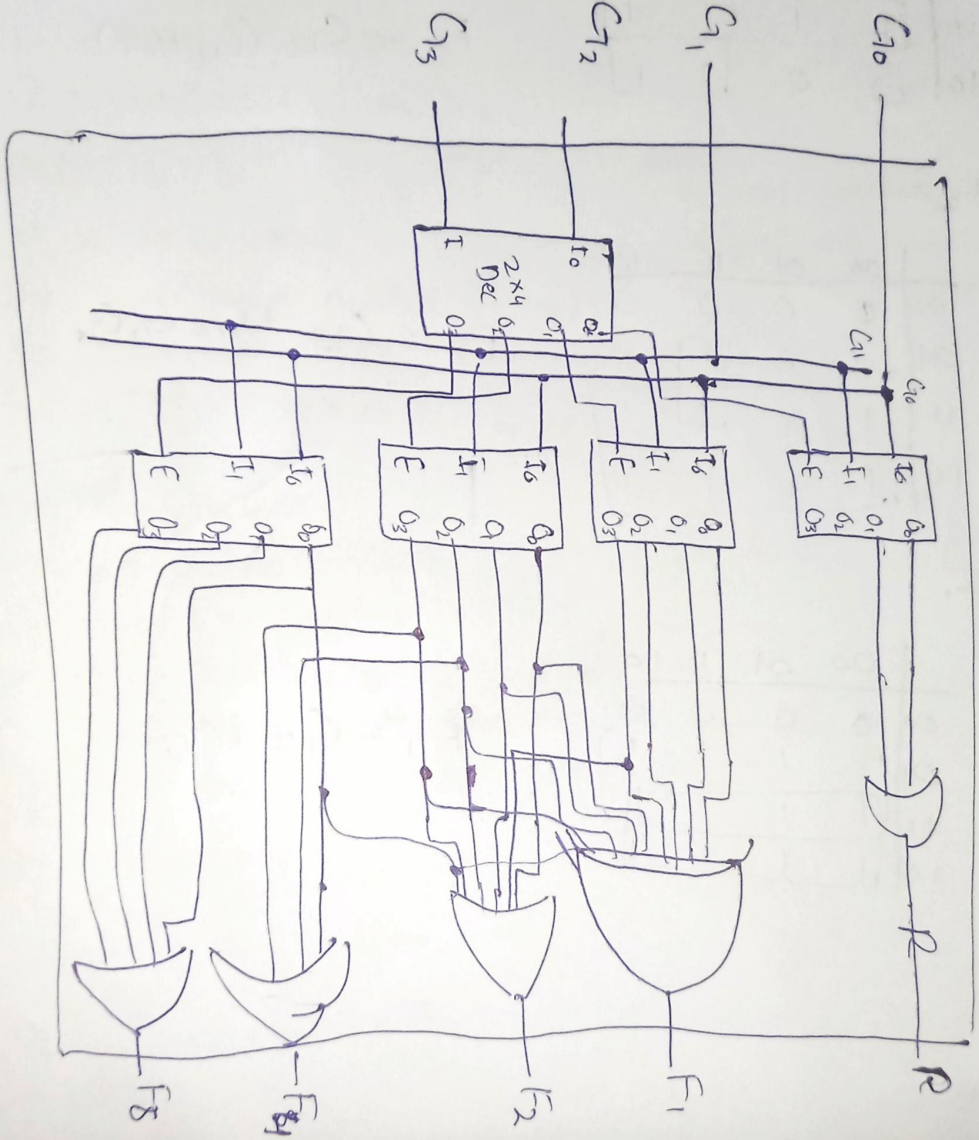
$F_1 = G_2 + G_3$

(5) R

	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$R = \overline{G_3}\,\overline{G_2}\,\overline{G_1}$

Sequential diagram:



Q No 3:

Design FSM that generate

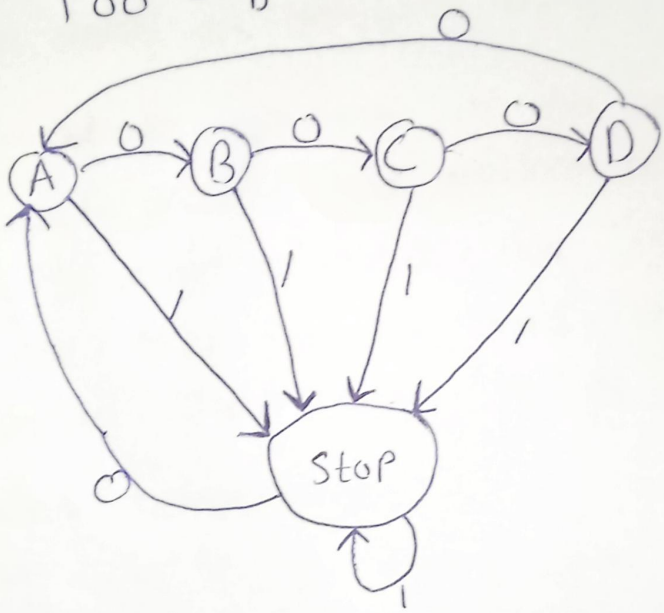
000 → 001 → 010 → 100 ↗ repeat

(a) S=1 → stop

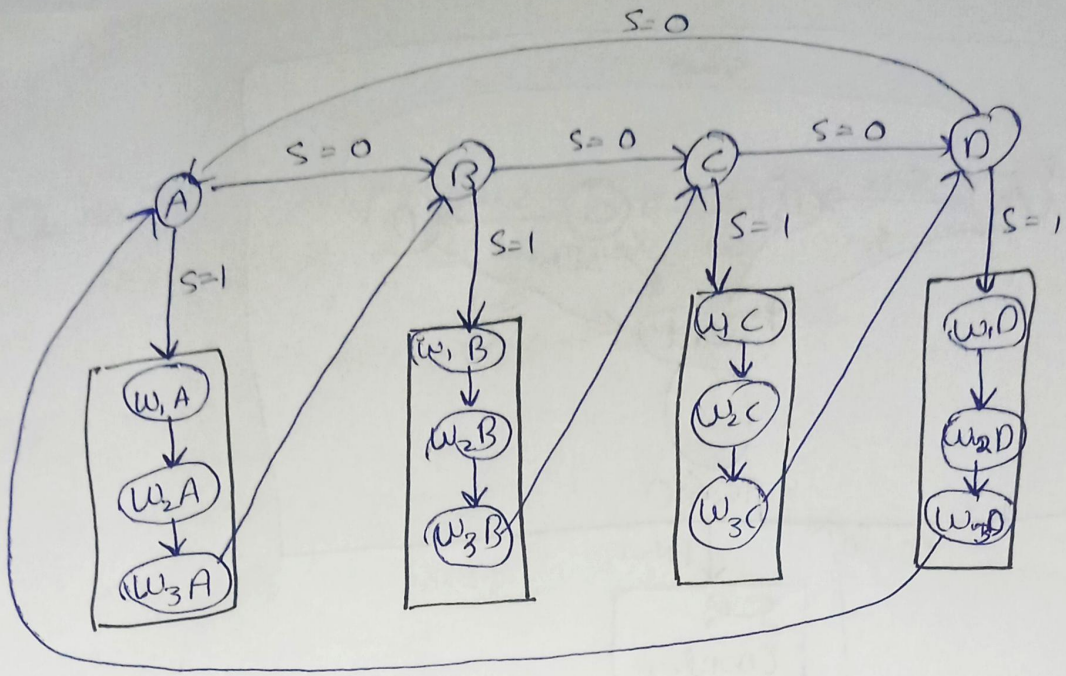
S=0 → restart from 000

let (State Encoding):

000 = A
001 = B
010 = C
100 = D



(b) S=1 → delay for 3 cycles, then continoues:



(c) S=1 → delay for 30 cycles using MSI counter

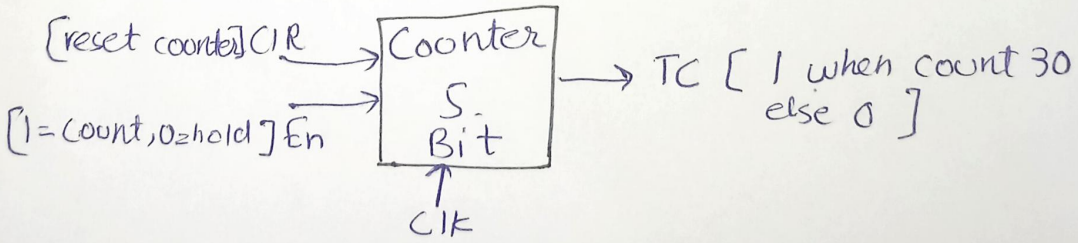
The above approch is to add new state for each wait that makes unnecessary states just to add a wait, but in this question we use MSI counter and integrate it with our fsm. That helps us to reduce states.

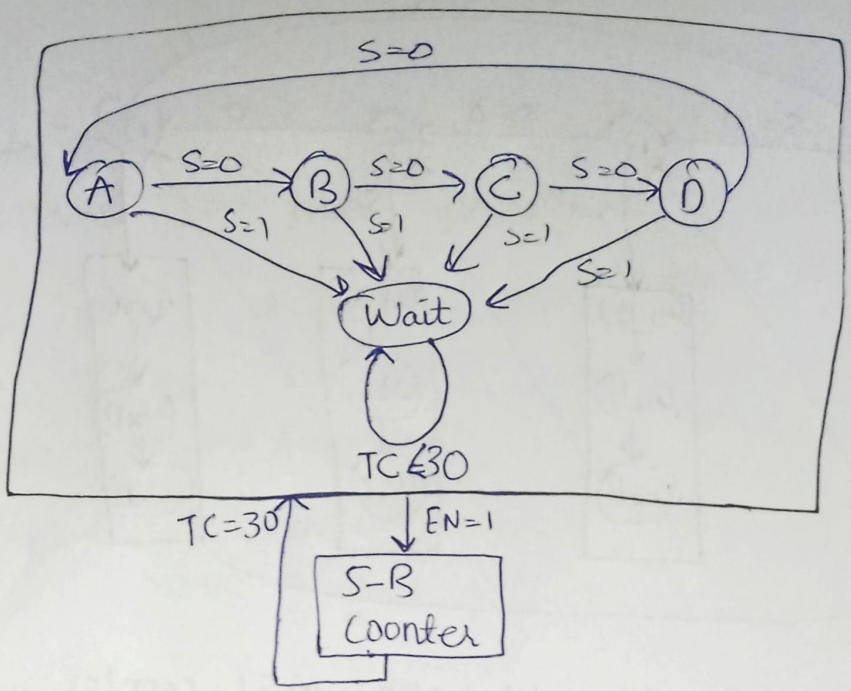
Counter component:

We have to count 30, so:

$$\log_2 30 \approx 5$$

We need 5 bit counter.




S=0
A
S=0
B
S=0
C
S=0
D
S=1
S=1
S=1
S=1
Wait
TC<30
TC=30
EN=1
5-B
Counter

Q NO 2:

Given:

Photo cost : 25 cent

Coins: Nickel(5), Dime(10), Quarter(25)

one coin / cycle

print photo : $\geq 25c$

return change and reset to initial state

Input:

N = Nickel (5 cent)

D = Dime (10 cent)

Q = Quarter (25 cent)

Output:

P = Print Photo

R_N = return nickel (5 cent)

R_D = return dime (10 cent)

R_T = return two dimes (20 cent)

Note:
State machine is mealey because we have to return output at multiple states

Required States:

(0 cent) S_0 : Initial State

(5 cent) S_1 : One nickel inserted

(10 cent) S_2 : Two nickel OR 1 dime

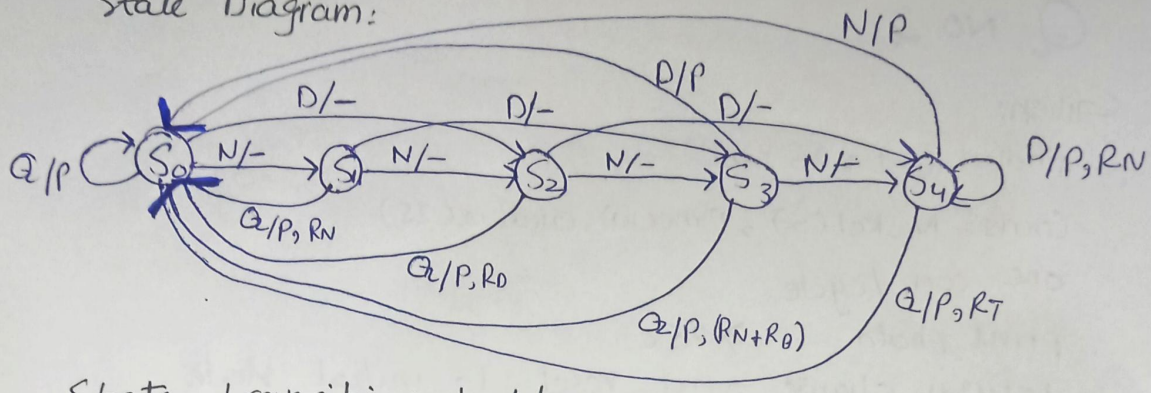
(15 cent) S_3 : three nickel OR nikel + dime

(20 cent) S_4 : four nickel OR two dime ...

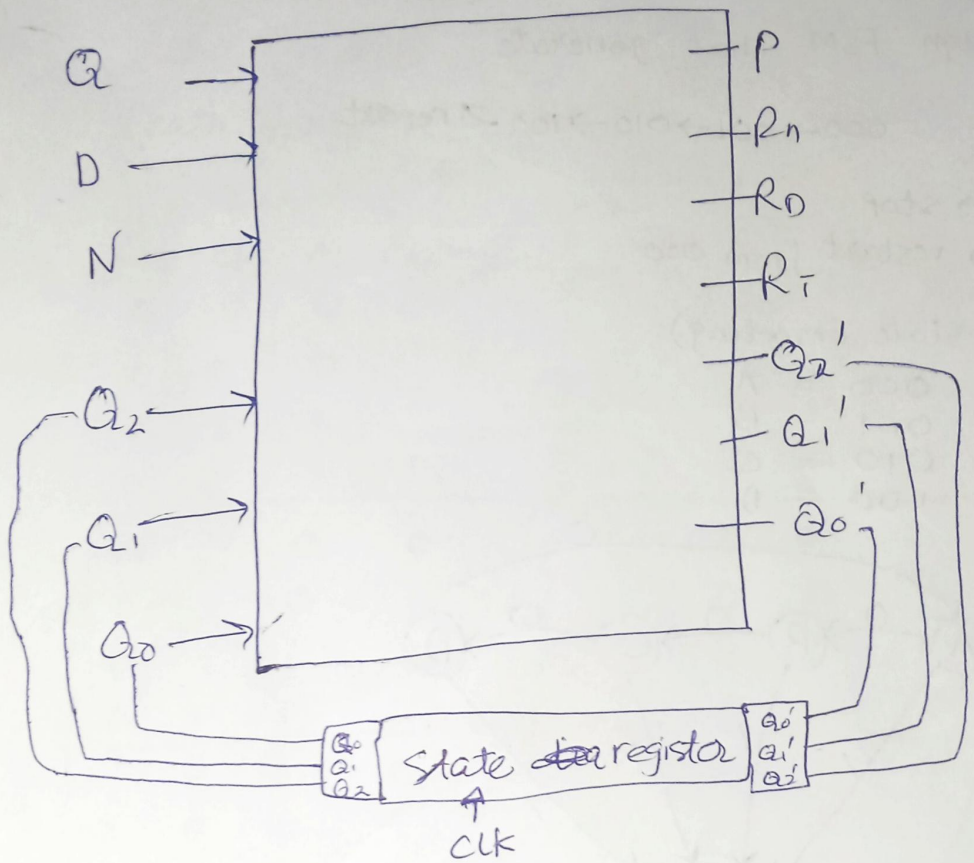
State Encoding:

	$Q_2Q_1Q_0$
S_0	000
S_1	001
S_2	010
S_3	011
S_4	100

State Diagram:



State transition table

Current State	Input	Next State	Output P	RN	RD	RT
$S_0(000)$	N	$S_1(001)$	0	0	0	0
$S_0(000)$	D	$S_2(010)$	0	0	0	0
$S_0(000)$	Q	$S_0(000)$	1	0	0	0
$S_1(001)$	N	$S_2(010)$	0	0	0	0
$S_1(001)$	D	$S_3(011)$	0	0	0	0
$S_1(001)$	Q	$S_0(000)$	1	1	0	0
$S_2(010)$	N	$S_3(011)$	0	0	0	0
$S_2(010)$	D	$S_4(100)$	0	0	0	0
$S_2(010)$	Q	$S_0(000)$	1	0	1	0
$S_3(011)$	N	$S_4(100)$	0	0	0	0
$S_3(011)$	D	$S_0(000)$	1	0	0	0
$S_3(011)$	Q	$S_0(000)$	1	1	1	0
$S_4(100)$	N	$S_0(000)$	1	0	0	0
$S_4(100)$	D	$S_0(000)$	1	1	0	0
$S_4(100)$	Q	$S_0(000)$	1	0	0	1

high level schematic: